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1 Overview

ResMed has developed a data platform to address the needs of ResMed partners seeking to integrate their own systems with ResMed AirView (formerly known as ResMed EasyCare Online).

AirView Exchange (AVX) (formerly known as ResMed EasyCare Exchange/ECX2) is a complete integration Application Programming Interface (API) suite designed for two-way communication from certified applications into the ResMed AirView system. This integration allows for workflow automation, reduced data entry effort and potential errors, as well as more timely and efficient access to the ResMed data. AVX is all about giving customers access to the data they need when they need it.

This document covers the Patient Management APIs exposed by AVX, which are intended for use by integration partners. It aligns with Release 4.38 of AirView for the Americas and the Asia-Pacific region.

2 AVX data management

AVX provides a number of data management services. These services are exposed as RESTful web APIs. This document explains how those APIs work and how an integrating application uses them.

2.1 Terminology

Throughout this specification, a number of terms are used. Refer to the table below for some of these key terms:

Key term	Description
API	Application Programming Interface
AVX	AirView Exchange
BCP	Best Current Practice
DLQ	Dead Letter Queue
DNS	Domain Name System
DNS Address	A symbolic name that resolves through the Domain Name System to an IP Address.
ECN	EasyCare Number. This number is a unique identifier that is used to identify a patient in AirView.
ECX2	EasyCare Exchange 2 (Now known as AirView Exchange/AVX)
IETF	Internet Engineering Task Force (a body that produces standards)
Integrating Application (IA)	An application (or system of applications) that connects to the AVX interfaces and uses the APIs. Many of the examples and explanations refer to this term. This application is supplied by the partners and connects to ResMed.
IP Address	Internet Protocol address of the system. In IPv4, this address would be <i>a.b.c.d</i> where <i>a</i> , <i>b</i> , <i>c</i> and <i>d</i> are decimal numbers from 0 to 255.
Integration User (IU)	When an Integrating Application connects to AVX APIs, it authenticates itself with the credentials of an Integration User. This user is the way that AVX determines what access is provided through the APIs. ResMed creates and manages this user account.
JSON	JavaScript Object Notation
PHI	Protected Health Information. Any information that can reasonably lead to the identity of a patient being recognized by someone who is not entitled to that knowledge.
REST	REpresentational State Transfer. A simple, stateless architecture that generally runs over HTTP.

Key term	Description
RESTful	A RESTful web API (also called a RESTful web service) is a web API implemented using HTTP and REST principles.
RSA	An asymmetric public, private key encryption algorithm.
Sandbox	A development and test environment.
SSL	Secure Sockets Layer
System	Generic term referring to an instance of a server built to meet these requirements.
TLS	Transport Layer Security. A cryptographic protocol used to provide secure communications over a computer network.
URI	Uniform Resource Identifier

2.2 Data managed via AVX

The ResMed AirView system stores a number of data elements and many of them can be accessed through the AVX API.

2.2.1 Patient records – demographics

AirView lets you create patient records and store all the pertinent information. This includes:

- Identification details such as first name, last name and date of birth.
- Prescription information, including the patient's therapy device and communication module.
- Physician associations.
- Clinical user and location associations.

Most of the fields entered into AirView can be both read and written from the AVX APIs.

This functionality can be used to achieve some important goals:

- Data is entered once in the IA and then loaded into AirView automatically.
- Via the AVX APIs, the same data is available in both the internal system and AirView – avoiding confusion, improving patient satisfaction and decreasing administration overhead.

2.3 Integration agreement

ResMed shares confidential information, grants access to resources and provides support to integrators. To ensure both parties are clear on expectations and requirements, this agreement defines the details. To integrate with ResMed systems, a current and valid integration agreement must be in place.

2.4 AVX development environment

ResMed will supply the following items in order for you to use the AVX development environment:

1. The sign in details for an AirView Administrator for a test organization.
2. The device details for some test therapy devices and communication modules that work with this test organization. These are simulated therapy devices, which can provide useful data for development and testing needs.
3. The connection details to access the AVX development APIs. The Integration User (IU) account used to connect to AVX is created in AirView using the Administrator account supplied in the first point above. This account is supplied in clear text format for development, but is encrypted in production.

3 Connectivity

3.1 Application connectivity to AVX

An integrating application connects to AVX using *incoming* RESTful web services. The term *incoming* is relative to AVX.

3.2 Network connection

The RESTful web service connections must be secured, as confidential patient information is communicated. The connections are also authenticated and the default authentication mechanism requires a secure channel for protection.

ResMed exposes the AVX RESTful web services to the internet as HTTPS connections (TLS/SSL Protected HTTP). The use of the TLS-encrypted connection for HTTP secures both the authentication phase and the data exchanged.

3.2.1 Data subscriptions and notifications

This (optional) API allows you to subscribe to get automatic notifications when AirView has new therapy session data for a patient.

AVX connects to your IA to send notifications using RESTful services (in a way that is similar to how your IA connects to AVX).

- To allow your IA to listen for notifications, you must create a RESTful service that has high uptime and availability.
- We recommend that you run the RESTful service independently of the primary IA to limit dependencies between the two systems that affect uptime and availability.

To connect from ResMed to the IA, the IA must:

1. *Expose the required RESTful web service(s) on the internet.*

The IA must communicate the URI for the RESTful endpoint to ResMed before AVX can send notifications. If this URI changes, the IA must also communicate those changes to ResMed.

When AVX tries to send a notification, it must be able to pass through any firewall or network configurations that you use in the IA environment. Therefore, it is important to consult with your network administrator before you create a RESTful service. Usually, network administrators configure their network with tighter controls for incoming connections than for outgoing connections. You may need to open up several network layers to enable access for incoming connections.

2. *Protect the connection using basic authentication.*

When ResMed sends a notification, it contains an "Authentication" header in the HTTP transmission. The IA must handle the basic type of authentication, and reject requests that do not contain an Authentication header or the correct credentials.

The basic authorization header contains a base64-encoded string that contains the colon-separated login name and password. It looks like the following:

Authorization: Basic QWxhZGRpbjpvYGVuIHNlc2FtZQ==

The base64 string is decoded to "Aladdin:open sesame" (without the quotation marks). The IA must parse the string and perform authentication.

3. *Secure the exposed RESTful web services using TLS/SSL certificates.*

To accept notifications, IA services must be able to connect over HTTPS.

You must use valid and verifiable certificates to respond to notifications. To ensure secure data transmission, ResMed only connects to services that are protected using

signed TLS/SSL certificates.

Note: Self-signed certificates are not suitable. It is recommended to use TLS v1.1 or greater.

While your RESTful web service must meet security, network and RESTful requirements, you can create the service using any underlying technologies or libraries.

3.3 Integration partner obligations

When an integration partner uses AVX, the IA is granted access to Protected Health Information (PHI) for the patients the partner cares for. This data is protected by the privacy legislation of the country the partner operates in. The partner should treat this connection and access to this connection with the same level of protection they apply to other systems in their organization that contain PHI. The partner is responsible for the organization controls required to protect the integrating application that uses AVX.

The IA stores the credentials that are used to access AirView. We recommend that the partner encrypts the AirView credentials to ensure they are secure. The integration partner must control access to the application. Since the IA has access to PHI contained within AirView, access to the IA must only be for treatment, payment for treatment, or health care operations.

The partner is responsible for monitoring access to the IA. In the event that a security incident is detected involving the IA, the partner must notify ResMed at privacy@resmed.com within 48 hours. ResMed will perform additional steps to protect the patients' data in AirView.

3.4 Limited number of requests for integration partners

AVX throttles the number of requests it processes to protect the AVX backend against severe traffic surges, which can lead to performance lags and downtime. If the number of requests reaches the set throttle limit, AVX returns a 429 HTTP response code (Too Many Requests). The integrating application can make another attempt after one second.

4 Security model

4.1 Adding an integrator to AirView/AVX (including authentication tokens)

When an IA is certified for integration through AVX, the integrator should provide the following information:

- Name as your company will be known within AirView.
- A public key.
- IP address or IP address range that you want to use for your production environment.
- (Optional) URL for notifications to be sent to.
- (Optional) Username and password for the notifications URL that uses basic authentication.

You are provided with an access token in the form of an encrypted username/passphrase.

The token is encrypted using PGP methodology with the public key registered in the AirView System when the IA was certified, along with AES encryption with a random key. The corresponding private key is known only to you and should be used on demand to obtain the information for connections.

Note: It is strongly recommended that the access token be stored in the encrypted form only in the IA and decrypted into memory when needed. This reduces the risk of someone obtaining the AVX Integration User (IU) account details and accessing the patient information without permission.

4.2 Process of providing access to patient information

As patients themselves do not currently log into AirView, the information about them is entered into AirView by a user of an AirView organization to whom the patient has given consent for their information to be stored.

Therefore, the AirView organization must explicitly grant a third party such as an Integrated Application (IA) access to the patient data. In AVX, that permission is granted by the AirView organization administrator authorizing the IA through the AirView user interface. They are identified within AirView by the name provided by the IA as per the previous section.

The IA's access token (used when calling the AVX APIs) allows access to that AirView organization. AVX enforces this restriction.

4.3 REST web service authentication

4.3.1 Integrating applications to AVX

Connections to AVX must be securely authenticated. The authentication process identifies the IU.

- The authentication process uses Basic Authentication (HTTP header) over SSL.
- The IU credentials used for the REST web service calls allow access to specific subsets of the ResMed Healthcare Informatics data and the associated APIs.
- The IU credentials provide access to AirView organizations as permitted by the organization's administrator through the AirView user interface.
- The IU credentials use a generated, secure passphrase that should not be exposed to any users.
- The IU credentials are provided to the IA upon addition of the IA to the production environment.

4.3.2 Protection from cross-site scripting (XSS) or malicious scripts

To protect AirView and its customers from malicious scripts, AVX rejects any API call that includes characters that are either invalid or specific to scripts, such as angle brackets ("**<**" or "**>**").

Additionally, if invalid characters or scripts reach the AirView database, AirView uses HTML encoding to protect AVX clients with the following symbol transformations:

- "&" symbol changes to "&"
- "<" symbol changes to "<"
- ">" symbol changes to ">"

While ResMed is taking a proactive approach to protect integrators and help avoid problems, ultimately the integrator is responsible for ensuring that the management and displaying of data is resistant to malicious data.

5 Uniform Resource Identifiers

The following table depicts the hierarchical structured design of AVX information. It also summarizes the HTTP calls that are available at each level.

URIs are specified in a manner that is consistent with RFC 6570 URI Template, and are case sensitive.

URI	Method	Description
http://<host>/healthCheck/status	GET	AVX health check. See "7.1 Health Check interface" on page 14.
https://<host>/v1/patients?device_serialNo={serial}&dob={date}	GET	Returns organization ID and ECN of matched patient. See "7.2.1 Search for patient by Device Serial No and Date of Birth (across all organizations)" on page 14.
https://<host>/v1/orgs	GET	Returns accessible organizations for the IA. See "7.2.2 Retrieve accessible organizations" on page 16.
https://<host>/v1/orgs/{orgID}/locations[?index={index}&count={count}]	GET	Returns all locations for the identified organization, paginated by index and count. See "7.2.3 Retrieve an organization's locations" on page 17.
https://<host>/v1/locations/{locationID}/clinicalUsers[?index={index}&count={count}]	GET	Returns all clinical users that have permission to access patients at the specified location of an organization. See "7.2.4 Retrieve clinical users who have access to a given location in an organization" on page 19.
https://<host>/v1/orgs/{orgID}/clinicalUsers[?index={index}&count={count}&clinicalUserID={clinicalUserID}]	GET	Returns all clinical users in an organization along with their details and locations to which they have access. See "7.2.5 Get all clinical users for an organization" on page 22.
https://<host>/v1/orgs/{orgID}/patients	GET	Returns a list of patient ECNs within the organization. See "7.2.6 Get patients for an organization" on page 25.
https://<host>/v1/orgs/{orgID}/integrators	GET	Returns the Patient Level Integrators for an organization. See "7.2.7 Get integrators for an organization" on page 26.
https://<host>/v1/orgs/{orgID}/availableBillingPlans[?cardOnly={value}]	GET	Returns the available billing plans for an organization. See "7.2.8 Get available billing plans for an organization" on page 28.

URI	Method	Description
https://<host>/v4/orgs/{orgID}/patients	POST	Used to create a new patient record for the specified organization. See "7.3.1 Create a new patient" on page 30.
https://<host>/v4/patients/{ecn}	PUT	Used to update a patient record. See "7.3.2 Update an existing patient" on page 34.
https://<host>/v2/patients/{ecn}	GET	Returns patient details for the requested ECN. See "7.3.3 Retrieve a patient's details" on page 39.
https://<host>/v1/patients/{ecn}/physicianOrgs	GET	Returns physician organizations and users linked to the patient ECN. See "7.3.4 Retrieve a patient's linked physicians" on page 43.
https://<host>/v1/patients/{ecn}/physicians/{id}	PUT	Used to add a physician to a patient. See "7.3.5 Add a physician to a patient" on page 45.
https://<host>/v1/patients/{ecn}/physicians/{id}	DELETE	Used to remove a physician from a patient. See "7.3.6 Remove a physician from a patient" on page 46.
https://<host>/v1/patients/{ecn}/integrators	GET	Returns Patient Level Integrators linked to a patient. See "7.3.7 Retrieve a patient's linked integrators" on page 47.
https://<host>/v1/patients/{ecn}/integrators	POST	Used to add a Patient Level Integrator to a patient. See "7.3.8 Add an integrator to a patient" on page 49.
https://<host>/v1/patients/{ecn}/integrators/{id}	DELETE	Used to remove a Patient Level Integrator from a patient. See "7.3.9 Remove an integrator from a patient" on page 51.
https://<host>/v1/patients/{ecn}/billingPlanAssignment	PUT	Used to assign a billing plan to a patient. See "7.3.10 Set a patient's billing plan" on page 52.
https://<host>/v1/patients/{ecn}/billingPlanAssignment	GET	Returns the current billing plan assigned to a patient. See "7.3.11 Get a patient's billing plan" on page 54.
https://<host>/v1/icians[?index={index}&count={count}&countries={countrycode},{countrycode}]	GET	Returns paginated list of physician details according to index, count and countries parameters. See "7.4.1 Get physician list" on page 57.

URI	Method	Description
https://<host>/v1/orgs/{orgID}/payors[?index={index}&count={count}]	GET	Returns a paginated list of payors according to index and count parameters. See "7.5.1 Get payor list" on page 61.

5.1 Deprecated URIs

ResMed is in process of removing the following URIs from AVX. Any integrators using these calls should use the most up-to-date version of the API listed in the table above.

URI	Method	Replaced with
https://<host>/v1/orgs/{orgID}/orgStructure https://<host>/v2/orgs/{orgID}/orgStructure	GET	https://<host>/v1/orgs/{orgID}/locations[?index={index}&count={count}] and https://<host>/v1/locations/{locationID}/clinicalUsers[?index={index}&count={count}]
https://<host>/v1/orgs/{orgID}/patients https://<host>/v2/orgs/{orgID}/patients https://<host>/v3/orgs/{orgID}/patients	POST	https://<host>/v4/orgs/{orgID}/patients
https://<host>/v1/patients/{ecn} https://<host>/v2/patients/{ecn} https://<host>/v3/patients/{ecn}	PUT	https://<host>/v4/patients/{ecn}
https://<host>/v1/patients/{ecn}	GET	https://<host>/v2/patients/{ecn}

5.2 Unsupported URIs

No previously supported URIs were removed as of this release.

6 JSON object design

6.1 JSON design: IA requests (PUT/POST)

This information allows you to create or update a patient record.

These objects vary depending on the request. Examples are provided for each call.

6.2 JSON design: AVX response to IA requests (GET/PUT/POST/DELETE)

Response information can be:

- information sent in response to an IA GET request, such as patient details or patient data
- confirmation messages in response to an Update (PUT, POST or DELETE)
- error messages sent in response to IA requests (GET, PUT, POST or DELETE).

Each API call is presented in this API specification with its JSON schema and an example response.

6.3 JSON design: AVX to IA notifications

These notifications inform the IA of a change to a patient record that they have subscribed to. For example, AVX sends a notification if new data arrives for a patient record, and it is expected that the IA would then make a GET call to request that data from AVX.

These objects contain the resource that has changed, and the date and time it changed.

6.4 API versioning

As shown in the "5 Uniform Resource Identifiers" on page 10 section, the first element of each URI is a version number. This version number allows us to create new versions of existing APIs in future releases, while continuing to support previous APIs for a defined period.

The versions are per API call; therefore, the most recent set of API calls at any point in time can consist of a variety of version numbers. Additionally, multiple versions of a call may exist concurrently. Older versions will be phased out to help maintain the interface methods. We will communicate these changes through an iteration of this technical specification.

Important note on versioning:

APIs using the GET method will not move to the next version if the change is only adding additional items to the body of the returned JSON object, and there is no impact to the structure/JSON path of the existing data items.

Additionally, POST/PUT APIs will not move to the next version if the change is only adding optional data items to the JSON object sent in the body of the POST.

- Integrating systems must be able to cope with additional items becoming part of a return for a GET and ignore them until such time that they are used by the integrating application.
- Integrating applications may also choose when to begin using additional optional objects in POST or PUT calls.

7 Interfaces

7.1 Health Check interface

The health check method allows integrators to determine the health of the AVX interface.

Request:

Property	Value
Method	GET
URL	http://<host>/healthCheck/status

Responses:

Success:

If the interface is available, an HTTP message is sent with an embedded JSON "Response" object containing the following properties:

Property	Value
HTTP Response Status Code	200
Content Type	application/json
Body	JSON "Response" object – See table below

Field Name (JSON Name)	Type	Description
responseCode	string	1000000
responseDesc	boolean	true
releaseVersion	string	Software version of AVX

Example:

```
{
  "HealthChecks": {
    "Response": {
      "responseCode": "1000000",
      "responseDesc": "true",
      "releaseVersion": "2.99.1-28"
    }
  }
}
```

Errors:

If the interface is unavailable, no response is sent.

7.2 Organization interfaces

7.2.1 Search for patient by Device Serial No and Date of Birth (across all organizations)

This API call is used to search for a patient by prescribed device serial number and date of birth. It is implemented across all organizations, whether accessible or not, and as such does not return any PHI.

If a patient with a matching device serial number and date of birth exists, the API returns the patient's ECN and organization ID. This allows the integrator to request more detailed information regarding this patient, provided they have access to that organization.

Request:

Property	Value
Method	GET
URL	https://<host>/v1/patients?device_serialNo={serial}&dob={date} where: <i>device_serialNo</i> is the device serial number being looked up. <i>dob</i> is the patient's date of birth that is being looked up. Format: YYYY-MM-DD.

Responses:

Success:

An HTTP 200 response with the following fields:

Property	Value
HTTP Response Status Code	200
Content Type	application/json
Body	JSON "Response" object – See table below

Refer to the JSON schema for information structure. Fields applicable to this API call are provided below.

Field Name (JSON Name)	Type	Description
responseCode	string	1003000
responseDesc	string	Unspecified – Debugging purposes only
searchResult ()	object	n/a
orgID	string	The organization ID that the patient belongs to
ecn	string	Patient unique identifier

Example:

```
{
  "Response": {
    "responseCode": "1003000",
    "responseDesc": "SUCCESS",
    "searchResult": {
      "orgID": "2",
      "ecn": "a0a2c85f-72c5-43ca-84ed-9a08e60caf3e"
    }
  }
}
```

Note: Data is fictional.

Errors/Not found:

Property	Value
HTTP Response Status Code	See table below
Content Type	application/json
Body	JSON "Response" object – See table below

Field Name (JSON Name)	Type	Description
responseCode	string	See table below
responseDesc	string	Unspecified – Debugging purposes only
fieldname	string	Any applicable field name – Debugging purposes only

Error Description	HTTP Response Status Code	responseCode
Patient not found	HTTP 200	1003001

7.2.2 Retrieve accessible organizations

This API call is used to retrieve the list of accessible organizations for the requesting integrator. An organization is accessible to an integrator if they have added the integrator as an integration partner through AirView. This can be useful in finding what orgIDs can be used when accessing other API calls.

Property	Value
Method	GET
URL	https://<host>/v1/orgs

Responses:

Success:

An HTTP 200 response with the following fields:

Property	Value
HTTP Response Status Code	200
Content Type	application/json
Body	JSON "Response" object – See table below

The following fields are included in the JSON "Response" object:

Field Name (JSON Name)	Type	Description
responseCode	string	1005000
responseDesc	string	Unspecified – Debugging purposes only
orgs ([])	object array	orgStructure object
name	string	The organization name
id	string	The organization ID

Example:

```
{
  "Response": {
    "responseCode": "1005000",
    "responseDesc": "SUCCESS",
    "orgs": [
      {
        "name": "French Basic HME",
        "id": "9"
      },
      {
        "name": "Large HME",
        "id": "2"
      }
    ]
  }
}
```

Note: Data is fictional.

Errors:

There are no error conditions specific to this API call.

7.2.3 Retrieve an organization's locations

This API call is used to retrieve a paginated list of active locations within an organization. Locations are sorted in ascending alphabetical order based on location name. From this list, count locations starting at position index are returned. This API can be useful when creating and/or editing patients using the patient details interfaces, as a valid location (and clinical user) are required. The clinical users in this case are provided by the following API: "7.2.4 Retrieve clinical users who have access to a given location in an organization" on page 19.

Note: Inactive locations are excluded from the returned results.

Property	Value
Method	GET
URL	<code>https://<host>/v1/orgs/{orgID}/locations/?index={index}&count={count}</code> where: <i>orgID</i> is the organization ID <i>index</i> (optional parameter) is the first location from the list that is returned in the JSON object <i>count</i> (optional parameter) is the number of locations from the list returned in the JSON object
Notes on Parameters	1. <code>index=0</code> refers to the first location. 2. If <code>index</code> is greater than or equal to the total number of results, no results are returned. 3. If no <code>index</code> is provided, or provided <code>index < 0</code>, <code>index=0</code> is used. 4. <code>count</code> refers to the number of results returned in the JSON object. 5. If no <code>count</code> is provided, or provided <code>count < 0</code> or provided <code>count</code> is greater than the max allowed, <code>count = max allowed count</code> is used. 6. max allowed count is currently set to 500.

Example of pagination:

The following examples illustrate how pagination works.

An organization has eight locations that, sorted alphabetically, are:

```
Air Liquide Medical MO
Bestmed IN
BSDS Center Co. WA
DIGI Medical AZ
DIGI Medical Inc CA
DIGI Medical Inc NE
DIGI Medical KY
DIGI Medical Ltd DC
```

Note: Data is fictional.

Example 1:

Using `index=0`, `count=3` in the query returns the following results:

```
Air Liquide Medical MO
Bestmed IN
BSDS Center Co. WA
```

Note: Data is fictional.

The first three locations are returned.

Example 2:

Using index=1, count=3 in the query returns the following results:

```
Bestmed IN
BSDS Center Co. WA
DIGI Medical AZ
```

Note: Data is fictional.

Locations two to four are returned.

Example 3:

Using index=2, count=10 in the query returns the following results:

```
BSDS Center Co. WA
DIGI Medical AZ
DIGI Medical Inc CA
DIGI Medical Inc NE
DIGI Medical KY
DIGI Medical Ltd DC
```

Note: Data is fictional.

Locations three to twelve are attempted to be returned. However, as the organization only has eight locations, locations three to eight are returned.

Responses:

Success:

An HTTP 200 response with the following fields:

Property	Value
HTTP Response Status Code	200
Content Type	application/json
Body	JSON "Response" object – See table below

The following fields are included in the successful JSON "Response" object:

Note: responseCode and responseDesc are NOT included in a successful response.

Field Name (JSON Name)	Type	Description
totalCount	string	The total number of active locations for this organization, irrespective of the current index and count values
locations ([])	object array	Paginated list of locations in an organization
name	string	The location name
id	string	The location identifier

Example:

```
{
  "Response": {
    "totalCount": "60",
    "locations": [
      {
        "name": "Air Liquide Medical and Sons PW",
        "id": "106"
      },
      {
        "name": "Air Liquide Medical Group WY",
        "id": "120"
      },
      {
        "name": "Air Liquide Medical Inc LA",
        "id": "82"
      },
      {
        "name": "Air Liquide Medical Inc ME",
        "id": "85"
      }
    ]
  }
}
```

Note: Data is fictional.

Errors:

Property	Value
HTTP Response Status Code	See table below
Content Type	application/json
Body	JSON "Response" object – See table below

Field Name (JSON Name)	Type	Description
responseCode	string	See table below
responseDesc	string	Unspecified – Debugging purposes only
fieldname	string	Any applicable field name – Debugging purposes only

Error Description	HTTP Response Status Code	responseCode
Requesting user does not have access to the organization	HTTP 401	1005001

7.2.4 Retrieve clinical users who have access to a given location in an organization

This API call is used to retrieve a paginated list of non-archived clinical users for a specified location within an organization. Clinical users are sorted in ascending alphabetical order, based on last name and then first name. From this list, *count* users starting at position *index* are returned. This API can be useful when creating and/or editing patients using the patient details interfaces, as a valid clinical user (and location) are required. The locations in this case are provided by the following API: "7.2.3 Retrieve an organization's locations" on page 17.

Note: Archived clinical users are excluded from the returned results.

Property	Value
Method	GET
URL	https://<host>/v1/locations/{locationID}/clinicalUsers[?index={index}&count={count}] where: <i>locationID</i> is the location ID <i>index</i> is the first clinical user from the list that is returned in the JSON object <i>count</i> is the number of clinical users from the list returned in the JSON object
Notes on Parameters	1. index=0 refers to the first clinical user. 2. If index is greater than or equal to the total number of results, no results are returned. 3. If no index is provided, or provided index <0, index=0 is used. 4. count refers to the number of results returned in the JSON object. 5. If no count is provided, or provided count <0 or provided count is greater than the max allowed, count = max allowed count is used. 6. max allowed count is currently set to 500. 7. locationID can be obtained by running the retrieve organization's location API as specified in "7.2.3 Retrieve an organization's locations" on page 17.

Example of pagination:

Refer to "7.2.3 Retrieve an organization's locations" on page 17.

Responses:

Success:

An HTTP 200 response with the following fields:

Property	Value
HTTP Response Status Code	200
Content Type	application/json
Body	JSON "Response" object – See table on the following page

The following fields are included in the successful JSON "Response" object:

Note: responseCode and responseDesc are NOT included in a successful response.

Field Name (JSON Name)	Type	Description
totalCount	string	The total number of non-archived users who can access the specified location, irrespective of the current index and count values
clinicalUsers ([[]])	object array	Paginated list of clinical users with access to a location in an organization
name	object	The clinical user's name
title	string	The clinical user's title
firstName	string	The clinical user's first name
lastName	string	The clinical user's last name
id	string	Unique identifier for clinical user

Example:

```
{
  "Response": {
    "totalCount": "3",
    "clinicalUsers": [
      {
        "name": {
          "firstName": "Prudence",
          "lastName": "1"
        },
        "id": "3DF1089D-96A0-4AC1-A0E1-F20CC931F75A"
      },
      {
        "name": {
          "title": "MR",
          "firstName": "Don",
          "lastName": "CenterManager"
        },
        "id": "9CCF3EF2-FBED-4058-89F9-36BF1073DAAC"
      },
      {
        "name": {
          "title": "MISS",
          "firstName": "Prudence",
          "lastName": "Lady"
        },
        "id": "3CC5A9CF-8C45-4160-ABBE-A372EC0BFF58"
      }
    ]
  }
}
```

Note: Identities and data are fictional.

Errors:

Property	Value
HTTP Response Status Code	See table below
Content Type	application/json
Body	JSON "Response" object – See table below

Field Name (JSON Name)	Type	Description
responseCode	string	See table on the following page
responseDesc	string	Unspecified – Debugging purposes only
fieldname	string	Any applicable field name – Debugging purposes only

Error Description	HTTP Response Status Code	responseCode
Supplied location ID does not exist, or requesting user does not have access to the organization to which the supplied location ID belongs.	HTTP 401	1005101

7.2.5 Get all clinical users for an organization

This API call is used to retrieve all clinical users in an organization. This API is useful when you require an entire list of clinical users in an organization. This API is not ideal for patient creation. To create patients, the clinical users can be retrieved using the API referenced in "7.2.4 Retrieve clinical users who have access to a given location in an organization" on page 19.

Note: All clinical users from the organization, including enabled, locked and archived clinical users, are returned.

Property	Value
Method	GET
URL	https://<host>/v1/orgs/{orgID}/clinicalUsers[?index={index}&count={count}&clinicalUserId={clinicalUserId}] where: <i>orgID (mandatory)</i> is the organization ID. <i>index (optional)</i> is the first clinical user from the list that is returned in the JSON object <i>count (optional)</i> is the number of clinical users from the list that is returned in the JSON object <i>clinicalUserId (optional)</i> is the clinical user id (GUID) that an integrator can use to search for a specific clinical user example: https://<host>/v1/orgs/123/clinicalUsers?index=0&count=5&clinicalUserId=56236DD8-8476-45E3-BF5F-D8B777ED46BB
Notes on Parameters	1. index=0 refers to the first clinical user. 2. If index is greater than or equal to the total number of results, no results are returned. 3. If no index is provided, or provided index <0, index=0 is used. 4. count refers to the number of results returned in the JSON object. 5. If no count is provided, or provided count <0 or provided count is greater than the max allowed, count = max allowed count is used. 6. max allowed count is currently set to 500.
Accept	Although not required, if this property is present, it must be compatible with "application/json" content type.

Responses:

Success:

An HTTP 200 response with the following fields:

Property	Value
HTTP Response Status Code	200
Content Type	application/json
Body	JSON "Response" object – See table below

The following fields will be included in the JSON response array element:

Field Name (JSON Name)	Type	Description
totalClinicalUserCount	string	The total number of clinical users including enabled, locked and archived
clinicalUsers ([])	object array	Object array containing multiple clinical users
id	string	Unique identifier for the clinical user
userName	string	User name for the clinical user
idNumber	string	The ID number for the clinical user that can be set manually in AirView
email	string	The clinical user's email
accountStatus	string	The clinical user's AirView account status. The status can be one of the following: • Enabled • Locked • Archived
name ()	object	The name of the clinical user
title	string	The clinical user's title
firstName	string	The clinical user's first name
lastName	string	The clinical user's last name
locations []	string array	Array of strings containing the location ids that the clinical user has access to
locationId	string	The location ID

Example:

```
{
  "Response": {
    "totalClinicalUserCount": "2",
    "clinicalUsers": [
      {
        "id": "87654321-11AA-22BB-33CC-1234567890AA",
        "userName": "JackAdminUser",
        "idNumber": "234155",
        "email": "jackadmin@example.com",
        "accountStatus": "ENABLED",
        "name": {
          "title": "MR",
          "firstName": "Jack",
          "lastName": "Admin"
        },
        "locations": [
          "456",
          "23423"
        ]
      },
      {
        "id": "87654321-11AA-22BB-33CC-1234567890AA",
        "userName": "JillAdminUser",
        "idNumber": "531235",
        "email": "jilladminuser@example.com",
        "accountStatus": "ARCHIVED",
        "name": {
          "firstName": "Jill",
          "lastName": "Admin"
        },
        "locations": [
          "789",
          "654"
        ]
      }
    ]
  }
}
```

Note: Identities and data are fictional.

Errors:

Property	Value
HTTP Response Status Code	See table below
Content Type	application/json
Body	JSON "Response" object – See table below

The following response code values may be returned:

Error Description	HTTP Response Status Code	responseCode
The client is not authorized to make this request.	HTTP 401	1005101
Invalid organization identifier or access to organization not allowed	HTTP 404	1005100
The requested clinical user was not found or does not exist.	HTTP 404	1005001

7.2.6 Get patients for an organization

This API call is used to retrieve a list of patient ECNs that belong to an organization. This can be useful to then allow further calls to be made on a patient. An optional date parameter can be specified to get a list of patient ECNs created after the specified date up to today's date.

Property	Value
Method	GET
URL	https://<host>/v1/orgs/{orgID}/patients[?fromDate={ISO8601Date}] where: <i>orgID</i> is the organization ID from which patients are retrieved <i>fromDate</i> specifies that only patients created after this date up to today's date will be returned
Notes on Parameters	1. fromDate is in the ISO 8601 combined date and time extended format yyyy-mm-ddThh:mm:ssZ (milliseconds can be specified but will be ignored). 2. If fromDate is not specified, then all patient ECNs in the organization will be retrieved.

Responses:

Success:

An HTTP 200 response with the following fields:

Property	Value
HTTP Response Status Code	200
Content Type	application/json
Body	JSON "Response" object – See table below

The following fields are included in the successful JSON "Response" object:

Field Name (JSON Name)	Type	Description
responseCode	string	1005000
responseDesc	string	Unspecified – Debugging purposes only
patients []	string array	A list of ECNs present within the organization

Example:

```
{
  "Response": {
    "responseCode": "1005000",
    "responseDesc": "SUCCESS",
    "patients": [
      "44187887-a48d-4f4e-9393-e4207045a2c4",
      "af3fb8e1-100c-4f84-84fb-1ca0adb8a2a2",
      "3cc5889d-4be4-433e-8101-a95c04d1a782",
      "61ed4b2a-fc2a-41cd-b42f-825fbef2da96"
    ]
  }
}
```

Note: Data is fictional.

Errors:

Property	Value
HTTP Response Status Code	See table below
Content Type	application/json
Body	JSON "Response" object – See table below

Field Name (JSON Name)	Type	Description
responseCode	string	See table below
responseDesc	string	Unspecified – Debugging purposes only
fieldname	string	Any applicable field name – Debugging purposes only

Error Description	HTTP Response Status Code	responseCode
Requesting user does not have access to the organization	HTTP 401	1005001
Bad date format	HTTP 400	1002100

7.2.7 Get integrators for an organization

This API call can be used to retrieve the Patient Level Integrators for an organization. The list of integrators is sorted in ascending order by the integrator name.

Property	Value
Method	GET
URL	https://<host>/v1/orgs/{orgID}/integrators where: <i>orgID</i> is the organization ID. example: https://<host>/v1/orgs/123/integrators

Responses:

Success:

Property	Value
HTTP Response Status Code	200
Content Type	application/json
Body	JSON "Response" object – See table below

The following fields are included in the JSON "Response" object:

Field Name (JSON Name)	Type	Description
responseCode	string	1002000
responseDesc	string	Unspecified – Debugging purposes only
orgIntegrators ([])	object array	Contains the organization's Patient Level Integrators.
id	string	The unique identifier of the integrator.
name	string	The integrator's name.

Example:

```
{
  "Response": {
    "responseCode": "1002000",
    "orgIntegrators": [
      {
        "id": "87654321-11AA-22BB-33CC-1234567890AA",
        "name": "Patient_Level_Integrator_1"
      },
      {
        "id": "87654321-11AA-22BB-33CC-1234567890BB",
        "name": "Patient_Level_Integrator_2"
      }
    ]
  }
}
```

Note: Data is fictional.

Errors:

Property	Value
HTTP Response Status Code	See table below
Content Type	application/json
Body	JSON "Response" object – See table below

The following fields are included in the JSON "Response" object:

Field Name (JSON Name)	Type	Description
responseCode	string	See table below
responseDesc	string	Unspecified – Debugging purposes only
fieldName	string	Any applicable field name – Debugging purposes only

The following response code values may be returned:

Error Description	HTTP Response Status Code	responseCode
The client is not authorized to make this request	HTTP 401	1005001
The requested organization was not accessible or was not found.		

Example:

```
{
  "Response": {
    "responseCode": "1005001"
  }
}
```

Note: Data is fictional.

7.2.8 Get available billing plans for an organization

This API call can be used to retrieve the available billing plans for an organization accessible by the IA.

A billing plan is considered available if the following conditions are met:

- the organization is permitted to use the billing plan
- the current date (as determined by the database server) is within the billing plan's availability date range.

Note: This is a date-only check and not a date/time check.

The available billing plans can be filtered by data access type.

Property	Value
Method	GET
URL	<code>https://<host>/v1/orgs/{orgID}/availableBillingPlans[?cardOnly={value}]</code> where: <i>orgID</i> is the organization ID. <i>cardOnly</i> indicates if the request is for billing plans with card-only data access (i.e. <i>cardOnly=true</i>), or billing plans with any data access type (i.e. <i>cardOnly=false</i>). This is optional. If this is not provided, then the default is billing plans with any data access type (i.e. <i>cardOnly=false</i>). example: <code>https://<host>/v1/orgs/123/availableBillingPlans</code> <code>https://<host>/v1/orgs/123/availableBillingPlans?cardOnly=true</code>

Responses:

Success:

Property	Value
HTTP Response Status Code	200
Content Type	application/json
Body	JSON "Response" object – See table below

The following fields are included in the JSON "Response" object:

Field Name (JSON Name)	Type	Description
responseCode	string	1002000
responseDesc	string	Unspecified – Debugging purposes only
availableBillingPlans ([])	object array	Contains the available billing plans.
id	string	The unique identifier of the billing plan.
name	string	The name of the billing plan.
availableStartDate	string	The date from which the billing plan is available (including the specified date) in YYYY-MM-DD format.
availableEndDate	string	The date to which the billing plan is available (including the specified date) in YYYY-MM-DD format. This is optionally pre-sent; it is not present if the billing plan is available indefinitely.

Example:

```
{
  "Response": {
    "responseCode": "1002000",
    "availableBillingPlans": [
      {
        "id": "87654321-11AA-22BB-33CC-1234567890AA",
        "name": "Card only",
        "availableStartDate": "2014-01-01"
      },
      {
        "id": "87654321-11AA-22BB-33CC-1234567890BB",
        "name": "Global Everything Plan",
        "availableStartDate": "2014-01-01",
        "availableEndDate": "2020-12-30"
      }
    ]
  }
}
```

Note: Data is fictional.

Errors:

Property	Value
HTTP Response Status Code	See table below
Content Type	application/json
Body	JSON "Response" object – See table below

The following fields are included in the JSON "Response" object:

Field Name (JSON Name)	Type	Description
responseCode	string	See table below
responseDesc	string	Unspecified – Debugging purposes only
fieldName	string	Any applicable field name – Debugging purposes only

The following response code values may be returned:

Error Description	HTTP Response Status Code	responseCode
• The client is not authorized to make this request.	HTTP 401	1005001
• The requested organization was not accessible or was not found.		

Example:

```
{
  "Response": {
    "responseCode": "1005001"
  }
}
```

Note: Data is fictional.

7.3 Patient interfaces

The following interface methods allow patients to be created or edited, or existing patient details to be retrieved.

7.3.1 Create a new patient

This API call is used to create a new patient in AirView in an organization that is accessible by the IA. This version supports S9™, Air10™, Air11™, Stellar™ and Astral™ devices only. Note that it is valid to create a patient without a device.

Property	Value
Method	POST
URL	https://<host>/v4/orgs/{orgID}/patients where: <i>orgID</i> is the organization ID within which the patient is to be created example: https://<host>/v4/orgs/123/patients
POST Content	A "patient" JSON object – See table below

The following table identifies the mandatory and optional fields contained within a "patient" JSON object for a patient create:

JSON Name	Type	Req'd	Rules/Format	Description
firstName	string	Yes	1 to 50 characters	Patient's first name
lastName	string	Yes	1 to 50 characters	Patient's last name
dob	string	Yes	YYYY-MM-DD	Patient's date of birth. The date cannot be in the future.
location	string	Yes	Valid location name in organization	The location name of the patient's assigned location. See "7.2.3 Retrieve an organization's locations" on page 17.
patientID	string	No	1 to 25 characters	Patient identifier. A customer-specified optional Patient ID field.
clinicalUser	string	Yes	Valid clinical user ID within organization	The user ID of the patient's assigned clinical user. See "7.2.4 Retrieve clinical users who have access to a given location in an organization" on page 19.
setupDate	string	Yes	YYYY-MM-DD	The date from which a patient's wireless monitoring schedule and compliance calculations begin in format YYYY-MM-DD. The date cannot be earlier than one year in the past or later than six months into the future.
physician	string	No	Valid physician user ID	The physician's user ID is included as an associated physician to the created patient. See "7.4.1 Get physician list" on page 57.

JSON Name	Type	Req'd	Rules/Format	Description
payorId	string	No	Valid payor ID	Payor ID to be associated with the patient. If it is not included, the organization's default payor is assigned. See "7.5.1 Get payor list" on page 61.
prescription ()	object	No	n/a	Prescription object
module ()	object	No ¹	n/a	Module object
serialNo	string	No	Valid module serial number allocated to organization and not currently assigned	Patient's communication module serial number
device ()	object	No	n/a	Device object
serialNo	string	No ²	Valid device serial number not currently assigned	Patient's prescribed device serial number
deviceType	string	No ²	See "8.1 S9 devices" on page 65	Patient's prescribed device type. Note: This is applicable for S9 devices only.
deviceNumber	string	No ³	Consists of 3 digits, from 0-9	Device number is required for Air10, Air11, Stellar, AirCurve™ 10 ST-A and Astral devices
mask ()	object	No	n/a	Patient's prescribed mask information
maskType	string	No ⁴	Valid predefined mask type. Case sensitive.	Mask type prescribed to patient. Refer to Appendix B: Mask Types and Sizes for the valid predefined mask types.
maskSize	string	No ⁴	Valid mask size for the specified predefined mask type. Case sensitive.	Mask size prescribed to patient. Refer to Appendix B: Mask Types and Sizes for the valid mask sizes.
otherType	string	No ⁴	1 to 255 characters	Mask type description when not a predefined mask type.
email	string	No	1 to 320 characters. Valid email address.	Patient's email address
gender	string	No	May be one of - MALE - FEMALE	Patient's gender
maritalStatus	string	No	May be one of - SINGLE - MARRIED - DIVORCED - WIDOWED	Patient's marital status
phoneNumbers ([])	object array	No	n/a	Phone numbers for the patient. Up to 5 may be supplied.

JSON Name	Type	Req'd	Rules/Format	Description
phoneNumber	string	No ⁵	1 to 35 characters. May contain characters other than numbers.	Patient's phone number
phoneType	string	No ⁵	May be one of - MOBILE - HOME - WORK - FAX - UNDEFINED	Phone number type

1 module() is applicable only for an S9 device, and must not be sent for other devices.

2 deviceType is required if a device is supplied and it is an S9. deviceType for Air10, Air11, Stellar, AirCurve 10 ST-A and Astral devices are known by AirView and as such must not be sent.

3 deviceNumber is required for Air10, Air11, Stellar, AirCurve 10 ST-A and Astral devices.

4 Mask information is optional. If you provide mask information, you must specify either (1) the maskType and maskSize for a predefined mask type or (2) otherType information.

5 Phone number information is optional. If you provide phone number information, you must specify the phoneNumber and phoneType for each phone number supplied.

Example (for S9 devices):

```
{
  "patient": {
    "dob": "1970-01-01",
    "firstName": "General",
    "lastName": "Hauck",
    "location": "Unimedis AL",
    "clinicalUser": "c501e66d-bcd0-4d99-b745-0970e65d55dd",
    "setupDate": "1990-01-01",
    "physician": "28a2898d-4e8e-45df-ae08-526337b74920",
    "prescription": {
      "module": {
        "serialNo": "09876543210"
      },
      "device": {
        "serialNo": "01234567890",
        "deviceType": "026001"
      },
      "mask": {
        "maskType": "mirage-softgel-nasal",
        "maskSize": "Large"
      }
    },
    "email": "general@example.com",
    "gender": "MALE",
    "maritalStatus": "SINGLE",
    "phoneNumbers": [
      {
        "phoneNumber": "+1 (555) 024-7777",
        "phoneType": "HOME"
      },
      {
        "phoneNumber": "61555867935",
        "phoneType": "UNDEFINED"
      }
    ]
  }
}
```

Note: Identities and data are fictional.

Example (for Air10, Air11, Stellar and Astral devices):

```
{
  "patient": {
    "dob": "1972-01-01",
    "firstName": "General",
    "lastName": "Williamson",
    "location": "Unimedis AP",
    "patientID": "HG656787",
    "clinicalUser": "28a2898d-4e8e-45df-ae08-526337b74920",
    "setupDate": "1999-01-01",
    "physician": "c501e66d-bcd0-4d99-b745-0970e65d55dd",
    "prescription": {
      "device": {
        "serialNo": "12547854521",
        "deviceNumber": "132"
      },
      "mask": {
        "maskType": "mirage-softgel-nasal",
        "maskSize": "Large"
      }
    },
    "email": "general@example.com",
    "gender": "MALE",
    "maritalStatus": "SINGLE",
    "phoneNumbers": [
      {
        "phoneNumber": "+1 (555) 024-7777",
        "phoneType": "HOME"
      },
      {
        "phoneNumber": "61555867935",
        "phoneType": "UNDEFINED"
      }
    ]
  }
}
```

Note: Identities and data are fictional.

Responses:

Success:

Property	Value
HTTP Response Status Code	200
Content Type	application/json
Body	JSON "Response" object – See table below

The following fields are included in the JSON "Response" object:

Field Name (JSON Name)	Type	Description
responseCode	string	1002000
responseDesc	string	Unspecified – Debugging purposes only
ecn	string	Unique identifier for created patient

Example:

```
{
  "Response": {
    "responseCode": "1002000",
    "responseDesc": "Patient Data Added",
    "ecn": "44187887-a48d-4f4e-9393-e4207045a2c4"
  }
}
```

Note: Data is fictional.

Errors:

Property	Value
HTTP Response Status Code	See table below
Content Type	application/json
Body	JSON "Response" object – See table below

Field Name (JSON Name)	Type	Description
responseCode	string	See table below
responseDesc	string	Unspecified – Debugging purposes only
fieldname	string	Any applicable field name – Debugging purposes only

Error Description	HTTP Response Status Code	responseCode
Mandatory Field Missing	HTTP 400	1002002
Invalid Parameter General	HTTP 400	1002100
JSON object does not match schema	HTTP 400	1002010
Device serial number already assigned to another patient	HTTP 403	1002101
Module serial number already assigned to another patient	HTTP 403	1002102
Module serial number not assigned to your organization	HTTP 403	1002103
Device type included without a device serial number	HTTP 400	1002100
Device serial number included without device type	HTTP 400	1002100
Incorrect serial number/device number combination	HTTP 403	1002105
Setup date shall not be earlier than one year prior or more than 6 months in the future	HTTP 400	1002106
DOB cannot be in the future	HTTP 400	1002107

7.3.2 Update an existing patient

This API call is used to update an existing patient record in AirView. The patient must be accessible by the IA.

This version supports S9, Air10, Air11, Stellar and Astral devices only. Note that it is valid for a patient to not have a device.

Property	Value
Method	PUT
URL	https://<host>/v4/patients/{ecn} where: <i>ecn</i> is the unique identifier of the patient being updated. example: https://<host>/v4/patients/44187887-a48d-4f4e-9393-e4207045a2c4
PUT Content	A "patient" JSON object – See table below

The following table identifies the mandatory and optional fields contained within a “patient” JSON object for a patient update. If a field is not included, it is preserved.

JSON Name	Type	Req'd	Rules/Format	Description
firstName	string	No	1 to 50 characters	Patient's first name
lastName	string	No	1 to 50 characters	Patient's last name
dob	string	No	YYYY-MM-DD	Patient's date of birth. The date cannot be in the future.
location	string	No	Valid location name in organization	The location name of the patient's assigned location. See "7.2.3 Retrieve an organization's locations" on page 17.
patientID	string	No	1 to 25 characters	Patient identifier
clinicalUser	string	No	Valid clinical user ID within organization	The user ID of the patient's assigned clinical user. See "7.2.4 Retrieve clinical users who have access to a given location in an organization" on page 19.
setupDate	string	No	YYYY-MM-DD	The date from which a patient's wireless monitoring schedule begins and from which compliance calculations begin with format YYYY-MM-DD. The date cannot be earlier than one year in the past or later than six months into the future.
physician	string	No	Valid physician user ID	The physician's user ID is included as an associated physician to the updated patient. See "7.4.1 Get physician list" on page 57.
payorId	string	No	Valid payor ID	Payor ID to be associated with the patient. If it is not included, the organization's default payor is assigned. See "7.5.1 Get payor list" on page 61.
prescription ()	object	No	n/a	Patient prescription information
module ()	object	No ¹	n/a	Module object
serialNo	string	No	Valid module serial number allocated to organization and not currently assigned	Patient's communications module serial number
device ()	object	No	n/a	Device object

JSON Name	Type	Req'd	Rules/Format	Description
serialNo	string	No ²	Valid device serial number not currently assigned	Patient's prescribed device serial number
deviceType	string	No ²	See "8.1 S9 devices" on page 65	Patient's prescribed device type. Note: This is applicable for S9 devices only.
deviceNumber	string	No ³	3-digit number	Device number is required for Air10, Air11, Stellar, AirCurve 10 ST-A and Astral devices.
mask ()	object	No	n/a	Patient's prescribed mask information
maskType	string	No ⁴	Valid predefined mask type. Case sensitive.	Mask type prescribed to patient. Refer to Appendix B: Mask Types and Sizes for the valid predefined mask types.
maskSize	string	No ⁴	Valid mask size for the specified predefined mask type. Case sensitive.	Mask size prescribed to patient. Refer to Appendix B: Mask Types and Sizes for the valid mask sizes.
otherType	string	No ⁴	1 to 255 characters	Mask type description when not a predefined mask type.
email	string	No	1 to 320 characters. Valid email address.	Patient's email address
gender	string	No	May be one of - MALE - FEMALE	Patient's gender
maritalStatus	string	No	May be one of - SINGLE - MARRIED - DIVORCED - WIDOWED	Patient's marital status
phoneNumbers ([])	object array	No	n/a	Phone numbers for the patient. Up to 5 may be supplied.
phoneNumber	string	No ⁵	1 to 35 characters. May contain characters other than numbers.	Patient's phone number
phoneType	string	No ⁵	May be one of - MOBILE - HOME - WORK - FAX - UNDEFINED	Phone number type

1 module() is applicable only for an S9 device, and must not be sent for other devices.

2 deviceType is required if a device is supplied and it is an S9. deviceType for Air10, Air11, Stellar, AirCurve 10 ST-A and Astral devices are known by AirView and as such must not be sent.

3 deviceNumber is required for Air10, Air11, Stellar, AirCurve 10 ST-A and Astral devices.

4 Mask information is optional. If you provide mask information, you must specify either (1) the maskType and maskSize for a predefined mask type or (2) otherType information.

5 Phone number information is optional. If you provide phone number information, you must specify the phoneNumber and phoneType for each phone number supplied.

Example (for S9):

```
{
  "patient": {
    "dob": "1970-01-01",
    "firstName": "General",
    "lastName": "Hauck",
    "location": "Unimedis AL",
    "clinicalUser": "28a2898d-4e8e-45df-ae08-526337b74920",
    "setupDate": "1990-01-01",
    "prescription": {
      "module": {
        "serialNo": "09876543210"
      },
      "device": {
        "serialNo": "01234567890",
        "deviceType": "026001"
      },
      "mask": {
        "maskType": "mirage-softgel-nasal",
        "maskSize": "Large"
      }
    },
    "email": "general@example.com",
    "gender": "MALE",
    "maritalStatus": "SINGLE",
    "phoneNumbers": [
      {
        "phoneNumber": "+1 (555) 024-7777",
        "phoneType": "HOME"
      },
      {
        "phoneNumber": "61555867935",
        "phoneType": "UNDEFINED"
      }
    ]
  }
}
```

Note: Identities and data are fictional.

Example (for Air10, Air11, Stellar and Astral devices):

```
{
  "patient": {
    "dob": "1972-01-01",
    "firstName": "General",
    "lastName": "Williamson",
    "location": "Unimedis AP",
    "patientID": "UTY766555",
    "clinicalUser": "c501e66d-bcd0-4d99-b745-0970e65d55dd",
    "setupDate": "1999-01-01",
    "prescription": {
      "device": {
        "serialNo": "12547854521",
        "deviceNumber": "132"
      },
      "mask": {
        "maskType": "mirage-softgel-nasal",
        "maskSize": "Large"
      }
    },
    "email": "general@example.com",
    "gender": "MALE",
    "maritalStatus": "SINGLE",
    "phoneNumbers": [
      {
        "phoneNumber": "+1 (555) 024-7777",
        "phoneType": "HOME"
      },
      {
        "phoneNumber": "61555867935",
        "phoneType": "UNDEFINED"
      }
    ]
  }
}
```

Note: Identities and data are fictional.

Responses:

Success:

Property	Value
HTTP Response Status Code	200
Content Type	application/json
Body	JSON "Response" object – See table below

The following fields are included in the JSON "Response" object:

Field Name (JSON Name)	Type	Description
responseCode	string	1002000
responseDesc	string	Unspecified – Debugging purposes only
ecn	string	Unique identifier for the updated patient

Example:

```
{
  "Response": {
    "responseCode": "1002000",
    "responseDesc": "Patient Data Updated",
    "ecn": "44187887-a48d-4f4e-9393-e4207045a2c4"
  }
}
```

Note: Data is fictional.

Errors:

Property	Value
HTTP Response Status Code	Function of error – See table below
Content Type	application/json
Body	JSON "Response" object – See table below

Field Name (JSON Name)	Type	Description
responseCode	string	Function of error – See table below
responseDesc	string	Unspecified – Debugging purposes only
fieldname	string	Any applicable field name – Debugging purposes only

Error Description	HTTP Response Status Code	responseCode
Requesting user does not have access to the patient	HTTP 401	1002001
Mandatory Field Missing	HTTP 400	1002002
JSON object does not match schema	HTTP 400	1002010
Invalid Parameter General	HTTP 400	1001003
Device serial number already assigned to another patient	HTTP 403	1002101
Module serial number already assigned to another patient	HTTP 403	1002102
Module serial number not assigned to your organization	HTTP 403	1002103
Device type included without a device serial number	HTTP 400	1002100
Device serial number included without device type	HTTP 400	1002100
Incorrect serial number/device number combination	HTTP 403	1002105
Setup date shall not be earlier than one year prior or more than 6 months in the future	HTTP 400	1002106
DOB cannot be in the future	HTTP 400	1002107

7.3.3 Retrieve a patient's details

This API call is used to obtain patient demographic data for a particular patient. The service is authenticated and only returns patients that are accessible by the IA.

Property	Value
Method	GET
URL	https://<host>/v2/patients/{ecn} where: <i>ecn</i> is the patient unique identifier. example: https://<host>/v2/patients/6fbbb12b-093e-4ab3-af9a-1071a13c4c02

Responses:

Success:

An HTTP 200 response with the following fields:

Property	Value
HTTP Response Status Code	200
Content Type	application/json
Body	JSON "Response" object – See table below

The following fields are included in the JSON "Response" object:

JSON Name	Type	Description	From Rev:
responseCode	string	1002000	2.2
responseDesc	string	Unspecified – Debugging purposes only	2.2
patient ()	object	Patient object	2.2
ecn	string	Patient unique identifier	2.2
patientID	string	Optional patient ID field	3.0
org ()	object	Organization object	2.2
name	string	Organization name	2.2
id	string	Organization ID that the patient belongs to	2.2
firstName	string	Patient's first name	2.2
lastName	string	Patient's last name	2.2
email	string	Patient's email address	3.5
dob	string	Patient's date of birth in YYYY-MM-DD format	2.2
location	string	Patient's assigned location	2.2
locationID	string	Patient's assigned location identifier (unique to the organization)	3.2
clinicalUser	string	The user ID of the patient's assigned clinical user. See "7.2.8 Get available billing plans for an organization" on page 28	4.13
setupDate	string	The date from which a patient's wireless monitoring schedule will begin and from which compliance calculation begins in YYYY-MM-DD format	2.2
insurers ([])	object array	Insurers for the patient	3.0
name	string	Name of insurer	3.0
membershipNumber	string	Patient's membership number	3.0
payorId	string	The ID of the payor associated with the patient	4.0
prescription ()	object	Patient prescription information	2.2
module ()	object	Patient's prescribed module information	2.2
serialNo	string	Patient's communication module serial number	2.2
device ()	object	Patient's prescribed device information	2.2
serialNo	string	The serial number of the device added to the patient's prescription	2.2
deviceType	string	A 6-digit unique identifier for the device prescribed to the patient	2.2

JSON Name	Type	Description	From Rev:
deviceTypeDesc	string	The name of the device prescribed to the patient	2.2
mask ()	object	Patient's prescribed mask information	3.0
maskType	string	Mask type prescribed to patient	3.0
maskSize	string	Mask size prescribed to patient	3.0
thresholds ()	object	The thresholds associated with this patient record	3.0
leak	string	The leak limit set for the patient in liters per second	3.0
gender	string	Patient's gender. May be one of the following: MALE FEMALE	4.13
myAirStatus	string	Patient's myAir™ invite status. May be one of the following: REGISTERED—user has registered in myAir NOT_REGISTERED—invite is not sent and user has not registered in myAir UNKNOWN—communication issue between AirView and myAir INVITED_NOT_REGISTERED—invite sent but user has still not registered in myAir	4.14
maritalStatus	string	Patient's marital status. Can be one of the following: SINGLE MARRIED DIVORCED WIDOWED	4.6
phoneNumbers ([])	object array	Phone numbers for the patient. Up to five may be returned.	4.6
phoneType	string	Phone number type. Can be one of the following: MOBILE HOME WORK FAX UNDEFINED	4.6
phoneNumber	string	Patient's phone number. May contain characters other than numbers.	4.6

Example:

```
{
  "Response": {
    "responseCode": "1002000",
    "responseDesc": "SUCCESS",
    "patient": {
      "org": {
        "name": "Large HME",
        "id": "2"
      },
      "clinicalUser": "3CC5A9CF-8C45-4160-ABBE-A372EC0BFF58",
      "prescription": {
        "device": {
          "serialNo": "87021125448",
          "deviceType": "026001",
          "deviceTypeDesc": "S9 AutoSet"
        },
        "mask": {
          "maskType": "mirage-softgel-nasal",
          "maskSize": "Large"
        }
      },
      "thresholds": {
        "leak": "10.2"
      },
      "insurers": [
        {
          "name": "insurance inc",
          "membershipNumber": "415"
        },
        {
          "name": "insurance corp",
          "membershipNumber": "413"
        }
      ],
      "dob": "1938-01-21",
      "ecn": "6fbbb12b-093e-4ab3-af9a-1071a13c4c02",
      "patientID": "541125",
      "firstName": "General",
      "lastName": "Hauck",
      "location": "Dream Corp AS",
      "locationID": "109",
      "setupDate": "2013-12-16",
      "payorId": "845",
      "status": "true",
      "email": "general@example.com",
      "gender": "MALE",
      "myAirStatus": "REGISTERED",
      "maritalStatus": "SINGLE",
      "phoneNumbers": [
        {
          "phoneNumber": "+1 (555) 024-7777",
          "phoneType": "HOME"
        },
        {
          "phoneNumber": "61555867935",
          "phoneType": "UNDEFINED"
        }
      ],
    }
  }
}
```

Note: Identities and data are fictional.

Errors:

Property	Value
HTTP Response Status Code	Function of error – See table below
Content Type	application/json
Body	JSON "Response" object – See table below

Field Name (JSON Name)	Type	Description
responseCode	string	Function of error – See table below
responseDesc	string	Unspecified – Debugging purposes only
fieldname	string	Any applicable field name – Debugging purposes only

Error Description	HTTP Response Status Code	Error Code
Requesting user does not have access to the patient	HTTP 401	1002001

7.3.4 Retrieve a patient's linked physicians

This API call can be used to obtain a list of physician accounts linked to the patient. The service is authenticated and can only return patients that are accessible by the IA.

Property	Value
Method	GET
URL	https://<host>/v1/patients/{ecn}/physicianOrgs where: <i>ecn</i> is the patient unique identifier.

Responses:

Success:

An HTTP 200 response with the following fields:

Property	Value
HTTP Response Status Code	200
Content Type	application/json
Body	JSON "Response" object – See table below

The following fields are included in the JSON "Response" object:

JSON Name	Type	Description	From Rev:
responseCode	string	1002000	4.6
responseDesc	string	Unspecified – Debugging purposes only	4.6
physicians([])	object array	Object array that contains information for one to many physicians in alphabetical order based on organization name and then physician name	4.6
id	string	Unique identifier for the physician	4.6
licenseNumber	string	Physician license number or NPI	4.6
org()	object	Object that contains organization details	4.6
name	string	Name of the physician organization that a patient is linked to	4.6
id	string	Unique identifier for the organization	4.6

JSON Name	Type	Description	From Rev:
name()	object	Object that contains physician name	4.6
title	string	Physician title	4.6
firstName	string	Physician first name	4.6
lastName	string	Physician last name	4.6

Note: An HME patient can be linked either to a physician within an organization, or directly to a physician organization. When a patient is linked directly to an organization the physician ID, licenseNumber, and name objects will not be returned. If a patient has no linked physicians then the entire physicians object will not be returned.

Example:

```
{
  "Response": {
    "responseCode": "1002000",
    "responseDesc": "SUCCESS",
    "physicians": [
      {
        "licenseNumber": "user20123",
        "id": "c9afd4da-c1fb-4f34-87d0-cdda26eea5c2",
        "org": {
          "name": "Air Liquide Medical and Sons",
          "id": "57"
        },
        "name": {
          "title": "Mr",
          "firstName": "Robin",
          "lastName": "Generated"
        }
      },
      {
        "org": {
          "name": "DIGI Medical",
          "id": "66"
        }
      }
    ]
  }
}
```

Note: Identities and data are fictional.

Errors:

Property	Value
HTTP Response Status Code	Function of error – See table below
Content Type	application/json
Body	JSON "Response" object – See table below

Field Name (JSON Name)	Type	Description
responseCode	string	Function of error – See table below
responseDesc	string	Unspecified – Debugging purposes only
fieldname	string	Any applicable field name – Debugging purposes only

Error Description	Response Type	Error Code
Requesting user does not have access to the patient	HTTP 401	1002001

7.3.5 Add a physician to a patient

This API call can be used to add a physician to a patient and currently only supports adding physician users and not organizations.

A request to add a physician that has already been added to the patient is considered a successful API call.

Property	Value
Method	PUT
URL	https://<host>/v1/patients/{ecn}/physicians/{id} where: <i>ecn</i> is the unique identifier of the patient. <i>id</i> is the unique identifier of the physician being added to the patient. example: https://<host>/v1/patients/12345678-11AA-22BB-33CC-1234567890AA/ physicians/87654321-11AA-22BB-33CC-1234567890AA

Responses:

Success:

Property	Value
HTTP Response Status Code	200
Content Type	application/json
Body	JSON "Response" object – See table below

The following fields are included in the JSON "Response" object:

Field Name (JSON Name)	Type	Description
responseCode	string	1002000
responseDesc	string	Unspecified – Debugging purposes only

Example:

```
{
  "Response": {
    "responseCode": "1002000"
  }
}
```

Note: Data is fictional.

Errors:

Property	Value
HTTP Response Status Code	See table below
Content Type	application/json
Body	JSON "Response" object – See table below

The following response code values may be returned:

Error Description	HTTP Response Status Code	responseCode
The client is not authorized to make this request.	HTTP 401	1002001
The requested patient was not accessible or was not found.		

Example:

```
{
  "Response": {
    "responseCode": "1002001"
  }
}
```

Note: Data is fictional.

7.3.6 Remove a physician from a patient

This API call can be used to remove a physician from a patient and currently only supports removing physician users and not organizations.

A request to remove a physician that has already been removed from the patient is considered a successful API call.

Property	Value
Method	DELETE
URL	<code>https://<host>/v1/patients/{ecn}/physicians/{id}</code> where: <i>ecn</i> is the unique identifier of the patient. <i>id</i> is the unique identifier of the physician being removed from the patient. example: <code>https://<host>/v1/patients/12345678-11AA-22BB-33CC-1234567890AA/physicians/87654321-11AA-22BB-33CC-1234567890AA</code>

Responses:

Success:

Property	Value
HTTP Response Status Code	200
Content Type	application/json
Body	JSON "Response" object – See table below

The following fields are included in the JSON "Response" object:

Field Name (JSON Name)	Type	Description
responseCode	string	1002000
responseDesc	string	Unspecified – Debugging purposes only

Example:

```
{
  "Response": {
    "responseCode": "1002000"
  }
}
```

Note: Data is fictional.

Errors:

Property	Value
HTTP Response Status Code	See table below
Content Type	application/json
Body	JSON "Response" object – See table below

The following fields are included in the JSON "Response" object:

Field Name (JSON Name)	Type	Description
responseCode	string	See table below
responseDesc	string	Unspecified – Debugging purposes only
fieldName	string	Any applicable field name – Debugging purposes only

The following response code values may be returned:

Error Description	HTTP Response Status Code	responseCode
The client is not authorized to make this request.	HTTP 401	1002001
The requested patient was not accessible or was not found.		

Example:

```
{
  "Response": {
    "responseCode": "1002001"
  }
}
```

Note: Data is fictional.

7.3.7 Retrieve a patient's linked integrators

This API call can be used to obtain a list of Patient Level Integrators linked to a patient accessible by the IA.

Property	Value
Method	GET
URL	https://<host>/v1/patients/{ecn}/integrators where: <i>ecn</i> is the unique identifier of the patient. example: https://<host>/v1/patients/12345678-11AA-22BB-33CC-1234567890AA/ integrators

Responses:

Success:

Property	Value
HTTP Response Status Code	200
Content Type	application/json
Body	JSON "Response" object – See table below

The following fields are included in the JSON "Response" object:

Field Name (JSON Name)	Type	Description
responseCode	string	1002000
responseDesc	string	Unspecified – Debugging purposes only
integrators ([])	object array	Contains the patient's Patient Level Integrators.
UID	string	The unique identifier of the integrator.
details	object	Contains the integrator details.
name	string	The integrator's name.
status	string	The integration status for this patient.

Example:

```
{
  "Response": {
    "responseCode": "1002000",
    "integrators": [
      {
        "UID": "87654321-11AA-22BB-33CC-1234567890AA",
        "details": {
          "name": "Patient_Level_Integrator_1",
          "status": "Incomplete"
        }
      },
      {
        "UID": "87654321-11AA-22BB-33CC-1234567890BB",
        "details": {
          "name": "Patient_Level_Integrator_2",
          "status": "Complete"
        }
      }
    ]
  }
}
```

Note: Data is fictional.

Errors:

Property	Value
HTTP Response Status Code	See table below
Content Type	application/json
Body	JSON "Response" object – See table below

The following fields are included in the JSON "Response" object:

Field Name (JSON Name)	Type	Description
responseCode	string	See table below
responseDesc	string	Unspecified – Debugging purposes only
fieldName	string	Any applicable field name – Debugging purposes only

The following response code values may be returned:

Error Description	HTTP Response Status Code	responseCode
• The client is not authorized to make this request.	HTTP 401	1002001
• The requested patient was not accessible or was not found.		

Example:

```
{
  "Response": {
    "responseCode": "1002001"
  }
}
```

Note: Data is fictional.

7.3.8 Add an integrator to a patient

This API call can be used to add a Patient Level Integrator to a patient.

A request to add an integrator that has already been added to the patient is considered a successful API call.

Property	Value
Method	POST
URL	https://<host>/v1/patients/{ecn}/integrators where: <i>ecn</i> is the unique identifier of the patient. example: https://<host>/v1/patients/12345678-11AA-22BB-33CC-1234567890AA/ integrators
POST Content	JSON "integrator" object – See table below
Content Type	Must contain the content-type "application/json".

The following table identifies the contents of the JSON "integrator" object:

Field Name (JSON Name)	Type	Req'd	Rules/Format	Description
integratorUid	string	Yes	Valid integrator ID	The unique identifier of the Patient Level Integrator being added to the patient.

Example:

```
{
  "integrator": {
    "integratorUid": "87654321-11AA-22BB-33CC-1234567890AA"
  }
}
```

Note: Data is fictional.

Responses:

Success:

Property	Value
HTTP Response Status Code	200
Content Type	application/json
Body	JSON "Response" object – See table below

The following fields are included in the JSON "Response" object:

Field Name (JSON Name)	Type	Description
responseCode	string	1002000
responseDesc	string	Unspecified – Debugging purposes only

Example:

```
{
  "Response": {
    "responseCode": "1002000"
  }
}
```

Note: Data is fictional.

Errors:

Property	Value
HTTP Response Status Code	See table below
Content Type	application/json
Body	JSON "Response" object – See table below

The following fields are included in the JSON "Response" object:

Field Name (JSON Name)	Type	Description
responseCode	string	See table below
responseDesc	string	Unspecified – Debugging purposes only
fieldName	string	Any applicable field name – Debugging purposes only

The following response code values may be returned:

Error Description	HTTP Response Status Code	responseCode
The requested entity was not valid JSON.	HTTP 400	1002010
The request entity was valid JSON but did not conform to the request schema defined above.	HTTP 400	1002001
- The client is not authorized to make this request. - The requested patient was not accessible to either the client or the requested integrator. - The requested patient was not found.	HTTP 401	1002001

Example:

```
{
  "Response": {
    "responseCode": "1002001"
  }
}
```

Note: Data is fictional.

7.3.9 Remove an integrator from a patient

This API call can be used to remove a Patient Level Integrator from a patient.

A request to remove an integrator that has already been removed from the patient is considered a successful API call.

Note: If an IA uses a Patient Level Integrator to make this API call, then the IA has permission to remove only that Patient Level Integrator from the patient.

Property	Value
Method	DELETE
URL	https://<host>/v1/patients/{ecn}/integrators/{id} where: <i>ecn</i> is the unique identifier of the patient. <i>id</i> is the unique identifier of the Patient Level Integrator being removed from the patient. example: https://<host>/v1/patients/12345678-11AA-22BB-33CC-1234567890AA/integrators/87654321-11AA-22BB-33CC-1234567890AA

Responses:

The following fields are included in the JSON "Response" object:

Field Name (JSON Name)	Type	Description
responseCode	string	1002000
responseDesc	string	Unspecified – Debugging purposes only

Example:

```
{
  "Response": {
    "responseCode": "1002000"
  }
}
```

Note: Data is fictional.

Errors:

Property	Value
HTTP Response Status Code	See table below
Content Type	application/json
Body	JSON "Response" object – See table below

The following fields are included in the JSON "Response" object:

Field Name (JSON Name)	Type	Description
responseCode	string	See table below
responseDesc	string	Unspecified – Debugging purposes only
fieldName	string	Any applicable field name – Debugging purposes only

The following response code values may be returned:

Error Description	HTTP Response Status Code	responseCode
- The client is not authorized to make this request. - The requested patient was not accessible or not found.	HTTP 401	1002001
The client is not permitted to remove the requested integrator from the patient.	HTTP 403	1002001

Example:

```
{
  "Response": {
    "responseCode": "1002001"
  }
}
```

Note: Data is fictional.

7.3.10 Set a patient's billing plan

This API call can be used to assign a billing plan to a patient whose organization is accessible by the IA.

You can assign a billing plan only if it is "available," which means that it meets the following conditions:

- the patient's organization is permitted to use the billing plan
- the current date (as determined by the database server) is within the billing plan's availability date range.

Note: This is a date-only check and not a date/time check.

Property	Value
Method	PUT
URL	https://<host>/v1/patients/{ecn}/billingPlanAssignment where: <i>ecn</i> is the unique identifier of the patient. example: https://<host>/v1/patients/12345678-11AA-22BB-33CC-1234567890AA/ billingPlanAssignment
POST Content	JSON "billingPlanAssignment" object – See table below

The following table identifies the contents of the JSON "billingPlanAssignment" object:

JSON Name	Type	Req'd	Rules/Format	Description
id	string	Yes	Valid billing plan ID	The unique identifier of the billing plan being assigned to the patient. The billing plan must be available before you can assign it to the patient.

Example:

```
{
  "billingPlanAssignment": {
    "id": "87654321-11AA-22BB-33CC-1234567890AA"
  }
}
```

Note: Data is fictional.

Responses:

Success:

Property	Value
HTTP Response Status Code	200
Content Type	application/json
Body	JSON "Response" object – See table below

The following fields are included in the JSON "Response" object:

Field Name (JSON Name)	Type	Description
responseCode	string	1002000
responseDesc	string	Unspecified – Debugging purposes only

Example:

```
{
  "Response": {
    "responseCode": "1002000"
  }
}
```

Note: Data is fictional.

Errors:

Property	Value
HTTP Response Status Code	See table below
Content Type	application/json
Body	JSON "Response" object – See table below

The following fields are included in the JSON "Response" object:

Field Name (JSON Name)	Type	Description
responseCode	string	See table below
responseDesc	string	Unspecified – Debugging purposes only
fieldName	string	Any applicable field name – Debugging purposes only

The following response code values may be returned:

Error Description	HTTP Response Status Code	responseCode
The request entity was not valid JSON.	HTTP 400	1002010

- The request entity was valid JSON but did not conform to the request schema defined above. - The patient does not have a device assigned to them. - The patient has an S9 device assigned to them. - The patient has an Air10 device without a communication module assigned to them. - The billing plan is not available to be assigned to the patient.	HTTP 400	1002001
- The client is not authorized to make this request. - The requested patient was not accessible or was not found.	HTTP 401	1002001

Example:

```
{
  "Response": {
    "responseCode": "1002001"
  }
}
```

Note: Data is fictional.

7.3.11 Get a patient's billing plan

This API call can be used to retrieve the current billing plan assigned to a patient accessible by the IA.

Property	Value
Method	GET
URL	https://<host>/v1/patients/{ecn}/billingPlanAssignment where: <i>ecn</i> is the unique identifier of the patient. example: https://<host>/v1/patients/12345678-11AA-22BB-33CC-1234567890AA/billingPlanAssignment

Responses:

Success:

Property	Value
HTTP Response Status Code	200
Content Type	application/json
Body	JSON "Response" object – See table below

The following fields are included in the JSON "Response" object:

Field Name (JSON Name)	Type	Description
responseCode	string	1002000

Field Name (JSON Name)	Type	Description
responseDesc	string	Unspecified – Debugging purposes only
billingPlan ()	object	Contains the billing plan details.
id	string	The unique identifier of the billing plan assigned to the patient.
name	string	The name of the billing plan assigned to the patient.

Example:

```
{
  "Response": {
    "responseCode": "1002000",
    "billingPlan": {
      "id": "87654321-11AA-22BB-33CC-1234567890AA",
      "name": "Card only"
    }
  }
}
```

Note: Data is fictional.

Errors:

Property	Value
HTTP Response Status Code	See table below
Content Type	application/json
Body	JSON "Response" object – See table below

The following fields are included in the JSON "Response" object:

Field Name (JSON Name)	Type	Description
responseCode	string	See table below
responseDesc	string	Unspecified – Debugging purposes only
fieldName	string	Any applicable field name – Debugging purposes only

The following response code values may be returned:

Error Description	HTTP Response Status Code	responseCode
- The client is not authorized to make this request. - The requested patient was not accessible or was not found.	HTTP 401	1002001
The requested patient does not have a billing plan currently assigned to them.	HTTP 404	1002001

Example:

```
{
  "Response": {
    "responseCode": "1002001"
  }
}
```

Note: Data is fictional.

7.4 Physician interface

7.4.1 Get physician list

Property	Value
Method	GET
URL	<code>https://<host>/v1/physicians[?index={index}&count={count}&countries={countrycode},{countrycode}]</code> where: <i>index</i> and <i>count</i> are optional query parameters that can be used to reduce the number of results retrieved. <i>index</i> is the starting point of the list. Note: If <i>index</i> is not provided, a value of 0 is used, which is the first record. If the <i>index</i> provided is greater than the number of physicians in AirView, no results are returned. <i>count</i> is the number of physicians to return in the call. <i>countries</i> is a query parameter that can be used to retrieve the list of physicians for one or more specified countries. If the country's query parameter is not provided, the full list of AirView physicians for this instance will be returned. Notes: The AirView configuration setting controls the optionality of the countries query parameter in specific regions (Americas, Asia-Pacific, Europe). By default, this query parameter is mandatory in Europe and optional in the Americas and Asia-Pacific. If the country's query parameter is not provided when a configuration setting is enabled, then an error is returned. For available country codes and their descriptions, see <i>Appendix C: Supported Country Codes</i>. Country codes are not case-sensitive. example: <code>https://<host>/v1/physicians</code> Will return a full list of physicians for the AirView instance (up to 10,000) when query parameters are not used to reduce the set. <code>https://<host>/v1/physicians?countries=FR</code> Will return the full list of AirView physicians for French organizations. <code>https://<host>/v1/physicians?countries=US,CA</code> Will return the full list of AirView physicians for US and Canadian organizations. Note: Only active physicians are returned by this API. The list of physicians returned is listed in alphabetical order by physician last name, then first name.

Responses:

Success:

Property	Value
HTTP Response Status Code	200
Content Type	application/json

Body	JSON object – See table below
------	-------------------------------

Note: This response does not include the response code and response description. This information is being phased out from success calls as it is redundant.

Field Name (JSON Name)	Type/Format	Description
totalPhysicianCount	string	Total number of physicians in AirView
physicians ([])	object array	Object array containing information for multiple physicians
id	string	Unique identifier for the physician
licenseNumber	string	Physician licence number or National Provider Identifier (NPI)
org ()	object	Object containing organization details
name	string	Name of the organization the physician belongs to
id	string	Unique identifier for the organization
name ()	object	Object containing physician name
title	string	Physician title
firstName	string	Physician first name
lastName	string	Physician last name
address ()	object	Object containing physician address
city	string	Physician city
state	string	Physician state
postCode	string	Physician postcode

Example:

```
{
  "totalPhysicianCount": "12345",
  "physicians": [
    {
      "licenseNumber": "8345456",
      "id": "c501e66d-bcd0-4d99-b745-0970e65d55dd",
      "org": {
        "name": "Large HME",
        "id": "2"
      },
      "name": {
        "title": "Dr",
        "firstName": "Fred",
        "lastName": "Frederickson"
      },
      "address": {
        "city": "Houston",
        "state": "tx",
        "postCode": "90210"
      }
    },
    {
      "licenseNumber": "2342458",
      "id": "85bab2ad-7b6d-4177-bb89-1733379a95e9",
      "org": {
        "name": "Large HME",
        "id": "2"
      },
      "name": {
        "title": "Dr",
        "firstName": "Sam",
        "lastName": "Anderson"
      },
      "address": {
        "city": "Los Angeles",
        "state": "ca",
        "postCode": "23451"
      }
    }
  ]
}
```

Note: Identities and data are fictional.

Errors:

Property	Value
HTTP Response Status Code	Function of error – See table below
Content Type	application/json
Body	JSON “Response” object – See table below

Field Name (JSON Name)	Type	Description
responseCode	string	Function of error – See table on the following page
responseDesc	string	Unspecified – Debugging purposes only
fieldName	string	Any applicable field name – Debugging purposes only

The following response code values may be returned:

Error Description	HTTP Response Status Code	responseCode
At least one country should be specified.	HTTP 400	1002002

• Provided country codes are invalid - [{countries}]	HTTP 400	1001003
• Provided country codes [{countries}] are not available in {region}		

Example:

```
{
  "Response": {
    "responseCode": "1002002",
    "responseDesc": "At least one country should be specified"
  }
}

{
  "Response": {
    "responseCode": "1001003",
    "responseDesc": "Provided country codes are invalid - [uk,ita]"
  }
}

{
  "Response": {
    "responseCode": "1001003",
    "responseDesc": "Provided country codes [us] are not available in Europe"
  }
}
```

Note: Data is fictional.

7.5 Payor interface

7.5.1 Get payor list

This API call can be used to retrieve a paginated list of payors associated with the specified organization. These payors can then be used to create and/or update a patient record. If a payor is not explicitly defined for a patient on create, the default payor is assigned.

Property	Value
Method	GET
URL	<code>https://<host>/v1/orgs/{orgID}/payors[index={index}&count={count}]</code> where: <i>index</i> and <i>count</i> are optional query parameters that can be used to reduce the number of results retrieved. <i>index</i> is the starting point of the list. Note: If <i>index</i> is not provided, a value of 0 is used, which is the first record. If <i>index</i> provided is greater than the number of payors in AirView, no results are returned. <i>count</i> is the number of payors to return in the call. Note: If <i>count</i> is not provided, the maximum AVX limit of 10,000 is used.

Responses:

Success:

Property	Value
HTTP Response Status Code	200
Content Type	application/json
Body	JSON object – See table below

The following fields are included in the JSON "Response" object:

Field Name (JSON Name)	Type/Format	Description
noOfPayors	string	Total number of payors associated with the organization
payors[]	object array	Object array containing information for multiple payors
name	string	Payor name
id	string	Unique identifier for the payor
transUnionId	string	The payor's identifier supplied by TransUnion®
isDefault	string	"Y" if this payor is the default for the organization "N" if it is not.

Example:

```
{
  "noOfPayors": "45",
  "payors": [
    {
      "name": "Access Medicare",
      "id": "845",
      "transUnionId": "11030",
      "isDefault": "Y",
    },
    {
      "name": "Alabama Medicaid",
      "id": "123",
      "transUnionId": "10007",
      "isDefault": "N",
    }
  ]
}
```

Note: Data is fictional.

Errors:

There are no error conditions specific to this API call.

7.6 Patient level integrator

7.6.1 Set patient level integrator status

This is only for integrating applications that have been set up as patient level integrator on individual patients. For this, the AirView user needs to follow the following steps:

1. The user logs in to AirView and selects a patient they want to integrate.
2. A PUT request is made to AVX to set the integration status to **Complete**.
3. The user is returned to the **Integrators** page using the referring URL.

Note: Your level of access to a patient is not changed based on their integration status (whether complete or incomplete).

Property	Value
Method	PUT
URL	<code>https://<host>/v1/patients/{ecn}/integrators/{integratorId}</code> (ResMed will provide this value.) where: <i>ecn</i> is the patient unique identifier <i>integratorId</i> is the unique identifier for the integrator.
PUT Content	An "integrator" JSON object – See table below

The following table identifies the mandatory fields contained within an “integrator” JSON object for an integration status change:

JSON Name	Type	Required	Rules/Format	Description
integrator ()	object	Yes	N/A	Integrator object
status	string	Yes	“complete” or “incomplete”	Sets the integrator status for the patient to be complete or incomplete. The status defaults to incomplete when a patient level integrator is assigned to a patient.

Example:

```
https://<avx>/exchange/v1/patients/{AirView.ecn}/integrators/{integratorId}
{
  "integrator":{
    "status":"complete"
  }
}
```

Responses:

Success:

An HTTP 200 response with the following fields:

Property	Value
HTTP Response Status Code	200
Content Type	application/json
Body	JSON "Response" object – See table below

The following fields are included in the JSON "Response" object:

JSON Name	Type	Description
responseCode	string	1002000

Example:

```
{
  "Response": {
    "responseCode": "1002000"
  }
}
```

Errors:

Property	Value
HTTP Response Status Code	Function of error – See table below
Content Type	application/json
Body	JSON "Response" object – See table below

Field name (JSON Name)	Type	Value
responseCode	string	Function of error – See table below
responseDesc	string	Unspecified – Debugging purposes only

Error Description	HTTP Response Status Code	Error code
The supplied integrator ID is incorrect	HTTP 401	1002001
The supplied ECN is incorrect or the integrator does not have access to the patient	HTTP 401	1002001

Error Description	HTTP Response Status Code	Error code
Supplied credentials are not correct	HTTP 401	1002001
The supplied JSON is malformed	HTTP 400	1002010
Supplied status is not "complete" or "incomplete"	HTTP 400	-

8 Appendix A. Supported devices

8.1 S9 devices

The following table includes the valid device types required to add an S9 device to a patient file.

S9 device name	deviceType
S9 AutoSet	026001
S9 Elite	026002
S9 Escape	026003
S9 Escape Auto	026004
S9 VPAP Auto	026009
VPAP S (S9)	026011
VPAP ST (S9)	026012
VPAP Adapt (S9)	026013
VPAP Adapt (S9 w/ ASVAuto)	026016
VPAP ST-A	026018
VPAP COPD	026023

8.2 All other devices

For devices manufactured after the S9, only the serial number and device number are used to add the device to a patient file.

For a list of all supported devices, see the Compatible devices topic in the AirView Help.

9 Appendix B. Mask types and sizes

Note: All masks that are available for selection in AirView are also supported in AVX. Certain mask types may not be available in some countries and regions.

Mask name as displayed in AirView	Mask type on AVX	Mask Size on AVX
AirFit F10	air-fit-f10	Small, Medium, Large
AirFit F10 for Her	air-fit-f10-for-her	X-Small, Small, Medium
AirFit F20	air-fit-f20	Small, Medium, Large
AirFit F20 for Her	air-fit-f20-for-her	Small, Medium
AirFit F30	air-fit-f30	Small, Medium
AirFit F30i	air-fit-f30i	Small, Medium, Small.Wide, Wide
AirFit N10	air-fit-n10	Standard, Wide
AirFit N10 for Her	air-fit-n10-for-her	Small
AirFit N20	air-fit-n20	Small, Medium, Large
AirFit N20 Classic	air-fit-n20-classic	Small, Medium, Large
AirFit N20 for Her	air-fit-n20-for-her	Small
AirFit N30	air-fit-n30	Small, Medium, Small.Wide
AirFit N30i	air-fit-n30i	Small, Medium, Small.Wide, Wide
AirFit P10	air-fit-p10	Small, Medium, Large
AirFit P10 for Her	air-fit-p10-for-her	X-Small, Small, Medium
AirFit P10 for AirMini	air-fit-p10-air-mini	X-Small, Small, Medium, Large
AirFit P30i	air-fit-p30i	Small, Medium, Large
AirTouch F20	air-touch-f20	Small, Medium, Large
AirTouch F20 for Her	air-touch-f20-for-her	Small, Medium
Mirage Activa LT	mirage-activa-lt-nasal	Small, Medium, Large, Large.Wide
Mirage Activa	mirage-activa-nasal	Standard, Shallow, Large
Mirage FX	mirage-fx	Standard, For.Her.Size, Wide
Mirage FX for Her	mirage-fx-for-her	Small, Standard
Mirage Liberty	mirage-liberty-full-face	Small, Large
Mirage Micro	mirage-micro-nasal	Small, Medium, Large, Large.Wide, X-Large
Mirage Quattro	mirage-quattro-full-face	X-Small, Small, Medium, Large
Mirage SoftGel	mirage-softgel-nasal	Small, Medium, Large, Large.Wide
Mirage Swift II	mirage-swift-ii-nasal-pillows	Small, Medium, Large
Mirage Swift	mirage-swift-nasal-pillows	Small, Medium, Large
Ultra Mirage	mirage-ultra-full-face	Small.Standard, Small.Shallow, Medium.Standard, Medium.Shallow, Large.Standard, Large.Shallow
Ultra Mirage II	mirage-ultra-ii-nasal	Standard, Shallow, Shallow.Wide, Large
Mirage Vista	mirage-vista-nasal	Standard, Deep
Pixi	pixi	One.Size
Quattro FX	quattro-fx	Small, Medium, Large
Swift FX	swift-fx	Small, Medium, Large
Swift FX for Her	swift-fx-for-her	X-Small, Small, Medium
Swift FX Bella	swift-fx-bella	X-Small, Small, Medium, Large

Mask name as displayed in AirView	Mask type on AVX	Mask Size on AVX
Swift LT for Her	swift-lt-for-her-nasal-pillows	X-Small, Small, Medium
Swift LT	swift-lt-nasal-pillows	Small, Medium, Large
Quattro Air	quattro-air	Small, Medium, Large
Quattro Air for Her	quattro-air-for-her	X-Small, Small, Medium
Quattro FX for Her	quattro-fx-for-her	Small, Medium, Large
Swift FX Nano	swift-fx-nano	Standard, Wide
Swift FX Nano for Her	swift-fx-nano-for-her	Small
Swift FX Bella Gray	swift-fx-bella-gray	Small, Medium, Large
NV Ultra Mirage	ultra-mirage-nv	Standard, Shallow, Large
NV Quattro FX	quattro-fx-nv	Small, Medium, Large
NV Quattro Air	quattro-air-nv	X-Small, Small, Medium, Large
AcuCare F1-0	acucare-f1-0	Small, Medium, Large
AcuCare F1-1	acucare-f1-1	Small, Medium, Large
AcuCare F1-4	acucare-f1-4	Small, Medium, Large
Mouthpiece	mouthpiece	Not-Applicable
Tracheostomy tube	tracheotomy	Not-Applicable
Cannula	cannula	Small, Medium, Large

10 Appendix C. Supported country codes

10.1 Americas country codes

Country code	Country name
US	USA
CA	Canada
AR	Argentina
EC	Ecuador
HN	Honduras
BO	Bolivia
DO	Dominican Republic
BR	Brazil
CL	Chile
CO	Colombia
CR	Costa Rica
CW	Curaçao
MX	Mexico
NI	Nicaragua
PA	Panama
PE	Peru
UY	Uruguay
VE	Venezuela

10.2 Asia-Pacific country codes

Country code	Country name
AU	Australia
JP	Japan
KR	South Korea



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